ARYA QUADRUPED ROBOT

ENGINEERING BILL OF MATERIALS — BOM

Project: ARYA Quadruped Robot **Document:** Engineering BOM

2. Robot Design Reference

Parameter	Specification
Robot Type	Quadruped Robot
Robot Height	**165 mm**
Robot Length	**250 mm**
Robot Width	**150 mm**
Leg Length	**120 mm**
Walking Speed	**60 mm/s**
Total Applied Load	**32.03 N**
Load per Leg	**≈8.01 N**
Static Friction Coefficient	**0.09**
Dynamic Friction Coefficient	**0.13**
Gait	Crawl Gait
Hip Servos	**4**

Knee Servos	**4**
Total Servos	**8**

3. BOM Classification

The BOM is divided into the following categories:

Category	Description
A	Electronics
B	Actuators
C	Servo Accessories
D	Battery & Power
E	3D-Printed Mechanical Components
F	Bearings
G	Standard Fasteners
H	Mounting Hardware
I	Cable Management
J	Foot / Contact Material
K	Manufacturing Consumables

4. Electronics BOM

Item No.	Component	Specifications	Dimensions	Qty	Unit Price	Tot Pri
E-01	ESP32 DevKit 30-Pin	ESP32-WROOM-32; 4 MB Flash; Wi-Fi 802.11 b/g/n; Bluetooth 4.2/BLE; 3.3 V logic	**≈52 × 28 × 4 mm**	1	₹519	₹5
E-02	PCA9685 PWM Driver	16 channels; 12-bit resolution; I²C; default address 0x40; 24 Hz–1.526 kHz	**≈62 × 25 × 3 mm**	1	₹249	₹2
E-03	GY-9250 / MPU9250 IMU	9-axis IMU; 3-axis accelerometer; 3-axis gyroscope; 3-axis magnetometer; I²C/SPI; 16-bit ADC	**≈25 × 15 mm**	1	₹365.80	₹365.8
E-04	OV2640 Camera Module	2 MP; 1600 × 1200; DVP 8-bit; SCCB; JPEG encoder	**≈10 × 10 × 20 mm** module	1	₹289	₹2

Electronics Known Subtotal

₹1,422.80

5. Actuator BOM

The quadruped uses **8 servo motors in total**:

- 4 Hip Servos
- 4 Knee Servos

5.1 Hip Servo Motors

Item No.	Component	Specifications	Dimensions	Qty	Unit Price	Total Price
A-01	Hip Servo	High-torque digital metal-gear servo; approximately 40 kg·cm class	**≈40 × 28 × 34 mm** target envelope	4	₹1,406	₹5,624

Supplier price basis: RoboticsDNA information supplied for the project.

5.2 Knee Servo Motors

Item No.	Component	Specifications	Dimensions	Qty	Unit Price	Total Price
A-02	Knee Servo	RDS3235 / CLS3335 class; high-torque digital servo; approximately 35 kg·cm class	**≈35 × 28 × 34 mm**	4	₹1,485	₹5,940

Actuator Subtotal

₹11,564

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6. Servo Accessories

Item No.	Component	Specifications	Dimensions	Qty	Unit Price	Tot Pri
A-03	Servo Horn	Aluminium; 25T spline; round/disc/arm type	Servo-specific; approximately Ø25–30 mm typical	8	₹27	₹2
A-04	Servo Mounting Screws	Servo-supplied mounting hardware; verify exact thread with selected servo	Servo-specific	8 sets	Included/ TBD	TE
A-05	Servo Extension Cable	3-wire servo power/signal extension	Length according to final wiring route	8	TBD	TE

Known Servo Accessory Subtotal

₹216

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7. Battery and Power BOM

Item No.	Component	Specifications	Dimensions	Qty	Unit Price	Total Price
P-01	3S LiPo Battery	2200 mAh; 11.1 V nominal; 12.6 V maximum; 30–40C continuous typical; JST-XH 4-pin balance connector	**≈110 × 40 × 28 mm** target	1	₹1,465	₹1,465
P-02	XT60 Battery Connector	Main battery power connector	Connector-specific; verify selected manufacturer	1 pair	TBD	TBD
P-03	Power Switch	Low-voltage DC rocker/ push switch	Vendor-specific	1	₹15	₹15
P-04	JST-XH 4-Pin Balance Extension	3S LiPo balance extension cable	Connector-specific	1	TBD	TBD

Known Power Subtotal

₹1,480

> **Battery verification:** The project specification gives approximately `110 × 40 × 28 mm`, while one supplier example supplied earlier was approximately `105 × 34 × 23 mm`. The final battery tray must

be checked against the actual purchased battery.

8. 3D-Printed Mechanical Components

> **Pricing is intentionally excluded for all components in this section.**

These parts must be manufactured using the approved CAD/STL files. Final dimensions should be taken from the released CAD revision.

Item No.	Component	Specifications / Function	Dimensions	Material	Qt
M-01	Upper Chassis / Main Body	Main structural body supporting legs and electronics	**250 × 150 mm footprint; thickness CAD-controlled**	Engineering-grade structural filament	
M-02	Lower Chassis / Electronics Plate	Supports electronics and battery	**250 × 150 mm footprint; thickness CAD-controlled**	Engineering-grade structural filament	
M-03	Hip Servo Bracket	Supports hip servo and maintains joint-axis alignment	Servo envelope ≈40 × 28 × 34 mm + clearance	Engineering-grade filament	

M-04	Thigh Link	Upper leg structural link	CAD-controlled; part of ≈120 mm leg assembly	Engineering-grade filament	
M-05	Shin / Lower Leg Link	Lower leg structural link	CAD-controlled; part of ≈120 mm leg assembly	Engineering-grade filament	
M-06	Battery Tray	Secures 3S 2200 mAh LiPo	**Internal clearance ≥110 × 40 × 28 mm**	Engineering-grade filament	
M-07	IMU Holder	Rigid mounting for MPU9250	**≈25 × 15 mm sensor envelope + clearance**	Engineering-grade filament	
M-08	Camera Mount	Holds OV2640 camera and maintains FOV	**≈10 × 10 × 20 mm envelope + clearance**	Engineering-grade filament	
M-09	Foot Connector	Mechanical interface between leg and foot	CAD-controlled	Engineering-grade filament	
M-10	Electronics Mounting Bosses	Provides PCB/ electronics mounting points	CAD-controlled	Engineering-grade filament	A require
M-11	Cable Routing Guides	Protects and guides wires around moving joints	CAD-controlled	Engineering-grade filament	A require

9. Mechanical Material Reference

Component Group	Material
Main load-bearing structure	Aluminium 6061 where specified by mechanical design
Mechanical links	Carbon Steel where specified by mechanical design
3D-printed prototype parts	Approved engineering-grade filament
Heat-set inserts	Brass
Servo gears	Hardened metal/steel depending on selected servo
Foot contact	Rubber
Servo mounting hardware	Steel
Bearings	Steel

10. Bearing BOM

Exact bearing selection must be finalized against the released CAD shaft and housing dimensions.

Item No.	Component	Specifications	Dimensions	Qty	Unit Price	Tota Pric

B-01	Joint Bearing	Deep-groove ball bearing; low-friction rotational support	**ID × OD × Width = CAD-controlled**	8	TBD	TB
B-02	Secondary Link Bearing	Bearing for secondary linkage if required by final CAD	**CAD-controlled**	As required	TBD	TB

Bearing Selection Requirements

The selected bearing must satisfy:

- Shaft diameter
- Housing diameter
- Bearing width
- Joint radial load
- Joint speed
- Proper fit
- Availability
- Serviceability

No bearing part number should be frozen until the final shaft and housing dimensions are confirmed.

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11. Standard Fasteners

11.1 Socket Head Cap Screws

Item No.	Component	Standard	Material / Grade	Dimensions	Qty	Unit Price
F-01	Socket Head Cap Screw	ISO 4762	Alloy Steel Grade 12.9	**M3 × length as per CAD**	As required	₹4.20

The supplied Fastdep Industrial price was approximately ₹**4.20 per piece**.

11.2 Hex Nuts

Item No.	Component	Standard	Dimensions	Qty	Unit Price	Total
F-02	Hex Nut	ISO 4032	**M3**	As required	TBD	TBD

11.3 Clevis Pins

Item No.	Component	Standard	Dimensions	Qty	Unit Price	Total
F-03	Clevis Pin with Head	ISO 2341	**M6 × length as per CAD**	8	TBD	TBD

11.4 Plain Washers

Item No.	Component	Standard	Dimensions	Qty	Unit Price	Total
F-04	Plain Washer	ISO 7089	**M6**	16	₹0.21	₹3.36

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12. Heat-Set Threaded Inserts

Item No.	Component	Specifications	Dimensions	Qty	Unit Price	Total
H-01	Heat-Set Threaded Insert	Brass insert for 3D-printed components	**M3 × insert length**	~20	TBD	TBD

Installation Requirements

- Compatible with selected filament
- Correct pilot-hole diameter
- Proper thermal installation
- Insert seated flush or to specified depth
- No cracking around insert
- Internal thread must accept M3 fastener correctly

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13. Electronics Mounting Hardware

Item No.	Component	Specifications	Dimensions	Qty	Unit Price	Tota
H-02	Nylon Standoff	PCB mounting standoff	**M3 × 10–20 mm**, final length CAD-controlled	8	TBD	TBI
H-03	Spacer	Nylon/metal electronics spacer	M3 compatible	As required	TBD	TBI
H-04	PCB Mounting Screw	Electronics mounting screw	M3 × required length	As required	TBD	TBI

14. Cable Management

Item No.	Component	Specifications	Dimensions	Qty	Unit Price	Total
C-01	Cable Tie	Nylon cable-management tie	**≈100–150 mm typical**	~20	TBD	TBD
C-02	Cable Clip	Servo/sensor wire routing clip	Small cable-compatible size	~10	TBD	TBD
C-03	Servo Extension Cable	3-wire servo extension	Length according to final leg routing	8	TBD	TBD

15. Rubber Foot Material

Item No.	Component	Specifications	Dimensions	Material	Qty	Unit Price
R-01	Rubber Foot Material	High-friction, non-slip ground-contact material	Thickness/ shape CAD-controlled	Rubber	4	TBD

Required Properties

- High friction
- Wear resistance
- Shock absorption
- Secure attachment
- Suitable for repeated ground contact

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16. Manufacturing Filament

Item No.	Component	Application	Specifications	Dimensions	Qty	
P-05	Engineering Nylon / Structural Filament	3D printing of structural parts	Engineering-grade filament	**Typically Ø1.75 mm if compatible with printer**	As per slicer	

Final quantity should be calculated from the validated STL files and slicer output.

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17. Complete Component Quantity Summary

Component	Quantity
ESP32 DevKit	1
PCA9685	1
MPU9250	1
OV2640 Camera	1
3S 2200 mAh LiPo	1
Hip Servo	**4**
Knee Servo	**4**
Total Servos	**8**
Servo Horn	**8**
Servo Extension Cable	**8**
XT60	**1 pair**
Power Switch	**1**
JST-XH Balance Extension	**1**
Upper Chassis	**1**
Lower Chassis	**1**
Hip Servo Bracket	**4**
Thigh Link	**4**
Shin Link	**4**
Foot Connector	**4**
Battery Tray	**1**

IMU Holder	**1**
Camera Mount	**1**
Joint Bearings	**8***
Heat-Set Inserts	**~20***
M3 SHCS	As required
M3 Hex Nuts	As required
M6 Clevis Pins	**~8***
M6 Washers	**~16***
PCB Standoffs	**~8***
Cable Ties	**~20***
Cable Clips	**~10***
Rubber Feet	**4**

`*` = planning quantity; final quantity must be reconciled against the released CAD assembly.

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18. Cost Summary

Only confirmed/supplied prices are included below.

Category	Known Cost
Electronics	₹1,422.80
Hip Servos	₹5,624
Knee Servos	₹5,940
Servo Horns	₹216

3S LiPo Battery	₹1,465
Power Switch	₹15
M6 Washers	₹3.36
Known Priced Subtotal	**₹14,686.16**

Excluded from known subtotal

The following require final supplier quotations:

- Bearings
- M3 screws
- M3 nuts
- Heat-set inserts
- Standoffs
- XT60
- JST-XH balance cable
- Servo extension cables
- Cable ties
- Cable clips
- Rubber foot material
- Filament
- 3D-printed parts
